

# CES Consumption Elasticity Estimation

## 1. Summary

This note documents the workflow used to update the estimation of CES substitution elasticities in the MAKRO consumption system. The work starts from raw MAKRO consumption quantities and prices, constructs Paasche-based nested consumption aggregates, calibrates CES share parameters, builds estimation-ready regression data, and estimates a sequence of increasingly rich models.

## 2. Folder structure

The workflow assumes the following working directory:

```
P:/leawin/Forbrugselasticitet
```

You have to change the working directory to another directory in all the codes to a folder on your own computer.

In this folder you have to create the following subfolders:

- Data/: CSV inputs and outputs from each estimation step.
- Tables/: PDF tables comparing sigma estimates across models.
- Plots/: residual and state-variable plots for diagnostics.
- R-koder/: R-scripts detailed below

## 3. Overview of scripts

| Script | Purpose                                                                                  | Main input                  | Main output                                           | Status             |
|--------|------------------------------------------------------------------------------------------|-----------------------------|-------------------------------------------------------|--------------------|
| 00     | Extract raw consumption quantities and prices from MAKRO/GDX.                            | makrobk.gdx                 | Data/forbrug_final.csv                                | Data extraction    |
| 01     | Create Paasche nest aggregates and calibrate omega for each nest.                        | Data/forbrug_final.csv      | Data/omega_nests_...csv;<br>Data/forbrug_nests_...csv | Nest construction  |
| 02     | Build one estimation-ready dataset with both inputs and observed nest aggregates.        | Omega and nest CSVs         | Data/regression_data_...csv                           | Regression data    |
| 03     | Estimate baseline joint system with fixed omega and linear preference trend.             | Data/regression_data_...csv | Baseline results, residuals, sigma table, plots       | Stable benchmark   |
| 04     | Estimate one free $\mu_t$ per year as a diagnostic preference component.                 | Data/regression_data_...csv | Flexible-mu results, residuals, mu plots              | Diagnostic         |
| 05     | Estimate structured $\mu_t = \tau_t + c_t$ without Kalman using penalty/state residuals. | Data/regression_data_...csv | Structured no-Kalman results, states, residuals       | Intermediate model |
| 06a    | Kalman ML using implied-mu measurement equation, fixed omega.                            | Data/regression_data_...csv | Kalman fixed-omega results, states, residuals         | Kalman variant     |

|     |                                                                                                        |                                 |                                                                                         |                                            |
|-----|--------------------------------------------------------------------------------------------------------|---------------------------------|-----------------------------------------------------------------------------------------|--------------------------------------------|
| 06b | Kalman ML with bounded free omega around calibrated omega.                                             | Data/regression_data_...csv     | Kalman bounded-free-omega results                                                       | Kalman variant                             |
| 06c | Direct-system Kalman using original relative demand and CES equations; separate measurement variances. | Data/regression_data_...csv     | Direct Kalman results, states, residuals                                                | Kalman variant                             |
| 06d | Direct-system Kalman with fixed omega and equal measurement variance across the two equations.         | Data/regression_data_...csv     | Direct equal-variance Kalman results, states, residuals                                 | Best Kalman candidate so far               |
| 07  | Compare old sigma values and all model estimates.                                                      | Result CSVs from 03-06d         | Data/comparison_all_sigma_results_...csv;<br>Tables/comparison_all_sigma_results_...pdf | Final comparison                           |
| 08  | Create final tables for the note                                                                       | Result CSVs from 03, 05 and 06d | Comparison tables                                                                       | Produces main result tables for the report |

## 00\_data\_fra\_makro.R

- Reads the MAKRO/GDX data source and extracts consumption quantities  $q_C$  and prices  $p_C$ .
- Merges quantities and prices by consumption component and year.
- Keeps the relevant consumption components used in the CES consumption nests.
- Exports the cleaned raw consumption dataset as Data/forbrug\_final.csv.

## 01\_make\_omega\_and\_nests.R

- Loads Data/forbrug\_final.csv and restricts the sample to the benchmark period, initially 1983-2017.
- Constructs Paasche price and quantity indexes for nested consumption aggregates.
- Defines the CES nest structure used in the old consumption paper.
- Calibrates one CES share parameter omega for each nest using the old sigma estimates and the Paasche-based data.
- Main outputs are Data/omega\_nests\_1983\_2017\_paasche.csv and Data/forbrug\_nests\_1983\_2017\_paasche.csv.

## 02\_make\_regression\_data.R

- Combines calibrated omega values with the Paasche nest data.
- For each nest, identifies the two inputs C1 and C2 and the observed aggregate  $C_{obs}$ .
- Adds prices, values, relative quantities, relative prices, and relative expenditure variables.
- Creates Data/regression\_data\_1983\_2017\_paasche.csv, which is the common input for all estimation scripts.

## 03\_estimate\_baseline\_fixed\_omega.R

- Estimates a first joint system consisting of a relative budget-share demand equation and a CES aggregation equation.
- Omega is kept fixed at its calibrated Paasche value.
- The preference term is simple:  $\mu_t = a + b * trend_t$ .
- This model produces stable sigma estimates close to the old paper and serves as the benchmark.
- Outputs include results\_fixed\_omega, residuals\_fixed\_omega, a comparison table with old sigma values, and residual plots.

## 04\_estimate\_mu\_flexible.R

- Replaces the linear preference trend with a flexible year-specific  $\mu_t$ .
- Estimates sigma and one free preference parameter for each year and nest.

- This is not intended as the final model; it checks whether a more flexible preference component changes sigma substantially.
- The sigma estimates remained close to the baseline, suggesting the baseline estimates were relatively robust to this diagnostic change.

#### 05\_estimate\_mu\_structured\_nokalman.R

- Introduces the structured preference decomposition  $\mu_t = \tau_t + c_t$ .
- Uses state-like restrictions  $\tau_t = \tau_{t-1} + \eta_t$  and  $c_t = \rho * c_{t-1} + \nu_t$ .
- This version does not use a Kalman filter. Instead, it estimates the system with nls.lm and adds state residual penalties.
- It is an intermediate non-Kalman approximation of the state-space model requested in the project.
- The sigma estimates again remained close to the baseline for most nests.

#### 06a\_estimate\_mu\_kalman\_fixed\_omega.R

- Implements the first full Kalman/maximum-likelihood system with fixed omega.
- Uses the state equations for  $\mu_t$ ,  $\tau_t$ , and  $c_t$ .
- The CES equation was initially transformed into an implied-mu observation.
- This version revealed identification problems: some sigma estimates moved to parameter bounds.

#### 06b\_estimate\_mu\_kalman\_free\_omega\_bounded.R

- Extends 06a by allowing omega to be estimated, but only within a bounded interval around the calibrated omega.
- This tests whether the CES share parameter can be estimated jointly with sigma and the state-space variances.
- The version improved likelihood in some nests but often pushed sigma and/or omega toward bounds, suggesting weak identification when omega is free.

#### 06c\_estimate\_mu\_kalman\_direct\_fixed\_omega.R

- Changes the Kalman implementation to use the original two measurement equations directly instead of an implied-mu transformation.
- The two measurement errors  $\text{var}(\epsilon_1)$  and  $\text{var}(\epsilon_2)$  are estimated freely.
- This version showed that the likelihood could put very different weights on the two equations, sometimes effectively down-weighting the CES equation.

#### 06d\_estimate\_mu\_kalman\_direct\_fixed\_omega\_equal\_measvar.R

- Uses the same direct-system Kalman setup as 06c, but imposes equal measurement variance across the relative demand and CES equations.
- Omega remains fixed at the calibrated Paasche value.
- This prevents the model from fully ignoring one of the two measurement equations by giving it a much larger error variance.
- This has been the best Kalman candidate so far: most sigma estimates are inside the bounds, although the energy-related nest still requires caution.

#### 07\_compare\_all\_sigma\_results.R

- Reads the sigma estimates from the old paper and all model variants.
- Builds a single comparison table across the old estimates, baseline, non-Kalman diagnostics, and Kalman variants.
- The purpose is to make it clear which models are stable and which estimates are driven by boundary or identification problems.
- Outputs a CSV and a PDF table for reporting.

08\_estimation\_table.R

- Create final tables for the note

#### 4. Running order

To reproduce the full workflow, run the scripts in this order:

- 00\_data\_fra\_makro.R
- 01\_make\_omega\_and\_nests.R
- 02\_make\_regression\_data.R
- 03\_estimate\_baseline\_fixed\_omega.R
- 04\_estimate\_mu\_flexible.R
- 05\_estimate\_mu\_structured\_nokalman.R
- 06a\_estimate\_mu\_kalman\_fixed\_omega.R
- 06b\_estimate\_mu\_kalman\_free\_omega\_bounded.R
- 06c\_estimate\_mu\_kalman\_direct\_fixed\_omega.R
- 06d\_estimate\_mu\_kalman\_direct\_fixed\_omega\_equal\_measvar.R
- 07\_compare\_all\_sigma\_results.R
- 08\_estimation\_table.R